Robustness of Synchronization of Heterogeneous Agents by Strong Coupling and a Large Number of Agents

Jaeyong Kim, Jongwook Yang, Hyungbo Shim, Jung-Su Kim, and Jin Heon Seo

***Abstract*—This technical note studies robustness of synchronization against heterogeneity of multi-agent systems. Here, heterogeneity also includes uncertainties and external disturbances in each agent. In order to effectively deal with the heterogeneous agents, we introduce the concept of the averaged dynamics which is the average of all agents' dynamics, and then claim that two sources enhance the robustness of synchronized behavior against heterogeneity of agents' dynamics. First, in spite of the heterogeneity of agents' dynamics, we show that strong coupling among agents makes the trajectories of all agents arbitrarily close to the averaged dynamics. Second, in the averaged dynamics, the effect of variations in each agent gets smaller as the number of agents increases. Simulation results confirm the claim that a large number of agents with strong couplings have robust synchronization.**

***Index Terms*— Averaged dynamics, heterogeneity, multi-agent systems, robustness, synchronization.**

I. Introduction

It is a common belief in biophysics and systems biology that synchronization makes the behavior of an interconnected system robust to perturbation, which has often been verified in simulations and experiments (see, e.g., [1], [2]). This technical note is devoted to explain how synchronization may help protect an interconnected multi-agent system from uncertainties and external disturbances. In fact, it is emphasized that the robustness comes from two specific components that lead to synchronization; that is, "multi"-agents and "strong coupling" among them. In particular, it is mathematically proved that (i) the solutions of individual agents get closer to each other as the coupling gain gets larger, so that practical synchronization is achieved, even under large heterogeneity among the agents, and (ii) as the number of agents becomes larger, the achieved synchronization becomes less affected by the variations in the individual agents. While the claim seems quite intuitive, its mathematical analysis for nonlinear dynamics has not been done until the work by Tabareau, Slotine, and Pham in [3], to the authors' knowledge.

Manuscript received May 15, 2015; revised October 23, 2015; accepted November 2, 2015. Date of publication November 5, 2015; date of current version September 23, 2016. This work was supported by a grant from the National Research Foundation of Korea (NRF), funded by the Ministry of Science, ICT and Future Planning (MSIP) (2009-0083495, 2015R1A2A2A01003878), and by the Human Resources Development of the Korea Institute of Energy Technology Evaluation and Planning (KETEP) grant funded by the Korea government Ministry of Trade, Industry & Energy (20154030200720). This paper was previously presented at the 2013 European Control Conference Zurich, Switzerland, July 2013. Recommended by Associate Editor C. Hadjicostis.

J. Kim, J. Yang, H. Shim, and J. H. Seo are with ASRI, Department of Electrical and Computer Engineering, Seoul National University, Seoul 151-744, South Korea (e-mail: hshim@snu.ac.kr).

J-S. Kim is with Department of Electrical and Information Engineering Seoul National University of Science and Technology, Seoul, 139-743, South Korea.

Color versions of one or more of the figures in this paper are available online at http://ieeexplore.ieee.org.

Digital Object Identifier 10.1109/TAC.2015.2498138

This technical note supports the same claim in a different setup. While identical agents under different external noises are considered in [3], we deal with heterogeneous (possibly uncertain and/or even unstable) agents represented by

$$\dot{x}_i = f_i(t, x_i) + u_i, \quad i \in \mathcal{N} := \{1, 2, \ldots, N\} \tag{1}$$

where N is the number of agents, $x_i \in \mathbb{R}$ is the state and $u_i \in \mathbb{R}$ represents interactions with other agents through the network. Here the function $f_i : [0, \infty) \times \mathbb{R} \to \mathbb{R}$ is uniformly bounded in t, continuously differentiable, and globally Lipschitz in x_i uniformly in t. Note that $f_i(t, x_i)$ may include time-varying signals persistently exciting the agent, and disturbances to the agent, as well as parametric variations or uncertainties of each agent. It is supposed that the agents are interconnected by static diffusive-type coupling [4]

$$u_i = k \sum_{j=1}^{N} \alpha_{ij}(x_j - x_i) \tag{2}$$

where k represents the coupling strength and α_{ij} is the (i, j)-th entry of the adjacency matrix of the given network.

Our handling of heterogeneity is enabled by introducing the notion of *averaged dynamics*, which is the average of the vector fields of each agent, given by

$$\dot{s}(t) = \frac{1}{N} \sum_{i=1}^{N} f_i\ (t, x_i(t))|_{x_i(t)=s(t)} =: \bar{f}\,(t, s(t)) \tag{3}$$

whose initial condition is set to the averaged one $s(0) = (1/N) \sum_{i=1}^{N} x_i(0)$. With the solution $s(t)$ of the averaged dynamics, it will be shown in Section II that, roughly stated, each solution $x_i(t)$ satisfies

$$\limsup_{t \to \infty} |x_i(t) - s(t)| \le \beta\left(\frac{1}{k}\right), \quad \forall\, k > \bar{K}, \quad \forall\, i \in \mathcal{N} \tag{4}$$

where $\bar{K}$ is a minimal required strength and β is a class-$\mathcal{K}$ function[1]—with the interpretation that "strongly coupled network makes each behavior of heterogeneous agent close to that of the averaged dynamics." This property will be validated under a certain stability (or, the contraction property [5]) of the averaged dynamics. It should be noted that the averaged dynamics can be stable even when some of the agents are unstable. It is true as long as the instability is compensated by the stability of other agents in the averaging process of (3). The averaged dynamics has often appeared in the literature (e.g., [6], [7], [18]). However, the analysis is mostly performed with the averaged trajectory $\bar{x}(t) := (1/N) \sum_{i=1}^{N} x_i(t)$ as a reference signal[2] rather than $s(t)$, where $x_i(t)$ is the solution to (1), (2). (Except at $t = 0$, $\bar{x}(t)$ and $s(t)$ are not the same in general.) Since $s(t)$ does not depend on the coupling gain k or on the network structure, it simplifies the analysis and shows the collective behavior of the multi-agent system with strong coupling before choosing k.

[1] A scalar continuous function $\gamma : [0, a) \to [0, \infty)$ is a class-$\mathcal{K}$ function if it is strictly increasing and $\gamma(0) = 0$.

[2] This also applies to non-smooth, but piecewise smooth, dynamics in [20].

0018-9286 © 2015 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See http://www.ieee.org/publications_standards/publications/rights/index.html for more information.
Authorized licensed use limited to: Gwangju Institute of Science and Technology. Downloaded on November 10,2026 at 18:09:19 UTC from IEEE Xplore. Restrictions apply.

On the other hand, we note that the averaged dynamics (3) is still affected by the variations of individual agents. For example, suppose that there is only one abnormal agent (i.e., having different vector field from others) in a group of identical and normal agents. If N is not large, the averaged dynamics may be affected heavily by the abnormal dynamics of the agent. Then, how to make the averaged dynamics robust to the perturbations in the abnormal agents? This is where the large number N comes into the picture. In Section III, only the parametric variations are dealt with for quantitative analysis, and the vector field $f_i(t, x_i)$ is supposed to be dependent on some parameters of random variables. With this setup, we additionally introduce the *expected averaged dynamics* $\dot{s}_E = E\{\bar{f}(t, s_E)\}$ as a reference system which is not affected by the variation of the agents, and show that, for any $\epsilon > 0$, the probability of the event that $\limsup_{t\to\infty} |s(t) - s_E(t)| \le \epsilon$ tends to 1 as $N \to \infty$;

$$\lim_{N\to\infty} P\left(\limsup_{t\to\infty} |s(t) - s_E(t)| \le \epsilon\right) = 1. \tag{5}$$

Combining these two results of (4) and (5), it is not difficult to infer that strong coupling and a large number of agents imply robustness of synchronization against heterogeneity (here, heterogeneity also includes uncertainty and/or external disturbance in each agent).

Consideration of heterogeneous agents in this way may explain one of the features of biological organs that they are meaningfully working well even though they consist of nonideal, imperfect, and fragile building blocks [8]–[10]. Indeed, the fact that any organ is composed of huge number of cells reduces the chance of ending up with abnormal averaged dynamics, and the fact that they are interconnected in a certain way (possibly with large coupling coefficients) implies the operations of the individual cells are not too different from others, and in this way, some malfunctioning cells can operate like normal ones within the network. If the lesson is transferred to another domain such as sensor networks [11], it becomes clear that one can enhance accuracy not by equipping with precision device but by employing many (cheap) sensors (which may include a few of defective ones) and combining their internal filters, such as Kalman filters, through the strong consensus network.

Now let us define some terminologies used throughout the technical note. An undirected graph is denoted by $\mathcal{G} = (\mathcal{N}, \mathcal{E}, \mathcal{A})$, where $\mathcal{N} = \{1, 2, \dots, N\}$ is a finite nonempty set of nodes, $\mathcal{E} \subseteq \mathcal{N} \times \mathcal{N}$ is an edge set of ordered pairs of nodes, and the *adjacency matrix* $\mathcal{A} = [\alpha_{ij}] \in \mathbb{R}^{N\times N}$ is defined such that $\alpha_{ji} = 1$ if $(i, j) \in \mathcal{E}$, while $\alpha_{ji} = 0$ if $(i, j) \notin \mathcal{E}$. The (symmetric) *Laplacian* matrix $\mathbb{L} = [l_{ij}] \in \mathbb{R}^{N\times N}$ of $\mathcal{G}$ is defined as $l_{ii} := \sum_{j\neq i} \alpha_{ij}$ and $l_{ij} := -\alpha_{ij}$ for all $i \neq j$. (We mean a is defined by b when we write $a := b$ or $b =: a$.) By its construction, it contains at least one eigenvalue of zero, whose corresponding eigenvector is 1_N ($N \times 1$ vector comprising all ones), and all the other eigenvalues are nonnegative. For a vector x and a matrix A, $|x|$ and $\|A\|$ denote the Euclidean norm and the induced matrix 2-norm, respectively. Let f and g be two functions on some subset of real numbers, $f(k) = \mathcal{O}(g(k))$ means that there exist constant c and K such that $f(k) \le cg(k)$ for all $k > K$.

II. Robustness by Strong Coupling

We study the problem under the following assumptions.

Assumption 1—(Individual Agent): The function $f_i(t, x_i)$ of the individual system (1) is uniformly bounded in t, continuously differentiable, and globally Lipschitz in x_i uniformly in t; i.e., there exist a non-decreasing continuous function $M : \mathbb{R}_{\ge 0} \to \mathbb{R}_{\ge 0}$ and a constant $L > 0$ such that, $\forall\, x_i \in \mathbb{R},\ \forall\, t \ge 0,\ \forall\, i \in \mathcal{N}$

$$|f_i(t, x_i)| \le M(|x_i|), \quad \left|\frac{\partial f_i}{\partial x_i}(t, x_i)\right| \le L. \tag{6}$$

Assumption 1 also guarantees uniqueness of the solutions $x_i(t)$, $\forall\, i \in \mathcal{N}$, for (1) and (2).

By letting $x := [x_1, \dots, x_N]^T$ and $f(t, x) := [f_1(t, x_1), \dots, f_N(t, x_N)]^T$, the dynamics of the overall system, composed of (1) and (2), is written as

$$\dot{x} = -k\mathbb{L}x + f(t, x) \tag{7}$$

where $\mathbb{L}$ is the Laplacian matrix describing the network connection.

Assumption 2—(Connected Network): The coupling network topology under consideration is undirected and connected.

A direct consequence of the assumption is that the Laplacian matrix $\mathbb{L}$ is symmetric and has zero eigenvalue which is simple [14], and it is supposed that, without loss of generality, the eigenvalues λ_i of $\mathbb{L}$ is such that $0 = \lambda_1 < \lambda_2 \le \cdots \le \lambda_N$. Therefore, by Schur's lemma, there exists a normal (real unitary) matrix $U \in \mathbb{R}^{N\times N}$ such that $U\mathbb{L}U^T = \mathrm{diag}(0, \Lambda)$ with a real diagonal matrix $\Lambda \in \mathbb{R}^{(N-1)\times(N-1)}$. Here we suppose that the first row of U is $(1/\sqrt{N})[1, 1, \dots, 1]$ (from the property of the Laplacian matrix). Define the matrix $W := (1/\sqrt{N})U$. Then

$$W = \begin{bmatrix} \frac{1}{N} 1_N^T \\ R^T \end{bmatrix}, \quad W^{-1} = \sqrt{N}U^T = [1_N, Q]$$

where R and Q are real matrices of size $N \times (N-1)$ such that $R^T R = (1/N)I$, $Q^T Q = NI$, $R^T 1_N = 0$, $Q^T 1_N = 0$, and $R^T Q = I$. Hence, $\|Q\| = \sqrt{N}$ and $\|R\| = 1/\sqrt{N}$.

Now, the *averaged dynamics* (or, so-called "blended dynamics") of (1) is defined as

$$\dot{s} = \frac{1}{N}\sum_{i=1}^{N} f_i(t, s) = \frac{1}{N} 1_N^T f(t, 1_N s) =: \bar{f}(t, s) \tag{8}$$

with the averaged initial condition $s(0) = \sum_{i=1}^{N} x_i(0)/N$. It is natural that $\bar{f}(t, s)$ is not the same as any of $f_i(t, s)$ so that $s(t)$ is different from any of $x_i(t)$. We assume stability of the averaged dynamics stated in the following way.

Assumption 3—(Stability of Averaged Dynamics): There exists a constant $p > 0$ such that, for all $s \in \mathbb{R}$ and $t \ge 0$

$$\frac{\partial \bar{f}}{\partial s}(t, s) = \frac{1}{N}\frac{\partial 1_N^T f}{\partial s}(t, 1_N s) = \frac{1}{N}\sum_{i=1}^{N} \frac{\partial f_i}{\partial x_i}(t, s) \le -p. \tag{9}$$

Note that this assumption does not require stability of individual systems, and allows unstable agents if their instability is compensated by the stability of other agents so that their sum is somehow stable in the sense of (9). A direct consequence of Assumption 3 is ultimate boundedness of the solution $s(t)$ of (8) as seen in the following lemma.

Lemma 1: For a scalar system $\dot{s} = F(t, s)$ with C^1 function F satisfying $(\partial F)/(\partial s) \le -p < 0$ for all s and $t \ge 0$

$$\limsup_{t\to\infty} |s(t)| \le \frac{\limsup_{t\to\infty} |F(t, 0)|}{p}. \tag{10}$$

♢

Proof: Since, by the mean-value theorem

$$\frac{d|s|}{dt} = \mathrm{sign}(s)\left(F(t, 0) + \left.\frac{\partial F(t, s)}{\partial s}\right|_{s=q} \cdot s\right)$$
$$\le |F(t, 0)| - p\,|s(t)|, \quad \text{almost everywhere}$$

with some $q \in \mathbb{R}$, the comparison lemma [12] yields $|s(t)| \le w(t)$, $\forall\, t \ge 0$, where $w(t)$ is the solution of $\dot{w} = -pw + |F(t, 0)|$ with $w(0) = |s(0)|$. This concludes the claim. □

Authorized licensed use limited to: Gwangju Institute of Science and Technology. Downloaded on November 10,2026 at 18:09:19 UTC from IEEE Xplore. Restrictions apply.

The following lemma, whose proof is found in the Appendix, will play a key role for the main result.

Lemma 2: Let

$$\rho_\kappa(x,y) = -\begin{bmatrix}|x|\\|y|\end{bmatrix}^T \begin{bmatrix}p & a\\ a & \kappa\end{bmatrix}\begin{bmatrix}|x|\\|y|\end{bmatrix} + \theta(t)|y|$$

with $x \in \mathbb{R}^l$, $y \in \mathbb{R}^m$, $p > 0$, $\theta(t) \geq 0$, and a is a constant. Then, there are a class-$\mathcal{K}$ function r and a positive number c such that

$$\rho_\kappa(x,y) \leq -c\left(|x|^2 + |y|^2\right) \quad \text{if } |x|^2 + |y|^2 > \theta^2(t) r\left(\frac{1}{\kappa}\right)$$

for all $\kappa > 3a^2/p$. ♢

Now one of the two main results in this technical note is presented in the following theorem, which shows strong coupling among agents makes the trajectories of all agents remain in an arbitrarily small neighborhood of the trajectory of the averaged dynamics [i.e., (4)].

Theorem 1: Under Assumptions 1, 2, and 3, there exists a class-$\mathcal{K}$ function γ_N such that the solutions of the overall system, composed of (1) and (2), with arbitrary initial conditions, and the solution $s(t)$ to the averaged dynamics (3) with $s(0) = \sum_{i=1}^N x_i(0)/N$ satisfy

$$\limsup_{t\to\infty} |x_i(t) - s(t)| \leq \gamma_N\left(\frac{1}{k\lambda_2 - L}\right), \quad \forall\, k > \bar{K} \tag{11}$$

for all $i = 1, \ldots, N$, where

$$\bar{K} = \frac{3L^2}{p\lambda_2} + \frac{L}{\lambda_2}. \tag{12}$$

In particular, the function γ_N is defined on $[0, p/3L^2)$ and given by

$$\gamma_N(\chi) = M\left(\frac{M(0)}{p}\right)\sqrt{N}\sqrt{r(\chi)} \tag{13}$$

in which

$$r(\chi) = \begin{cases} 0, & \chi = 0\\ \frac{4\chi}{p - 3L^2\chi}, & 0 < \chi \leq \frac{4p}{p^2+20L^2}\\ \frac{(p^2+8L^2)\chi^2}{(p-3L^2\chi)^2}, & \frac{4p}{p^2+20L^2} < \chi < \frac{p}{3L^2}. \end{cases} \tag{14}$$

Remark 1: The ultimate bound in (11) expressed by the function γ_N of (13), and the value of $\bar{K}$ of (12) may be conservative. However, the expressions (12) and (13) yield a reasonable interpretation. For example, (12) indicates that the minimal coupling strength $\bar{K}$ increases as the Lipschitz constant L increases while it decreases as the degree of stability p and the smallest non-zero eigenvalue λ_2 of the network get larger.

Remark 2: It should be noted that the function γ_N and the value of $\bar{K}$ are affected by the number N. The former is obvious due to the appearance of $\sqrt{N}$ in (13), but the latter is indirect through the value of λ_2. The second smallest eigenvalue λ_2 of the Laplacian matrix is called the algebraic connectivity (or, density) of a graph $\mathcal{G}$, which indicates how well connected the graph is. It depends both on the topology of the graph and the number N of the nodes. For the all-to-all network (with unit weights), λ_2 is the same as the number N, but for the ring network, increasing N decreases λ_2 (because $\lambda_2 = 2(1 - \cos(2\pi/N))$). We refer to [13] for more details on their relation.

Remark 3: Remark 2 implies that, in order to maintain the same level of error while the number N increases, the coupling strength k may need to be increased. In fact, the ultimate error bound is given in (11) and it can be seen that $\gamma_N(1/(k\lambda_2 - L)) = \mathcal{O}(\sqrt{N/k\lambda_2})$ when k is large enough (so that χ is small enough). Therefore, for the ring network where $\lambda_2 = 2(1 - \cos(2\pi/N))$, we have $\gamma_N = \mathcal{O}(\sqrt{N^3/k})$, and for the same level of error, k should be increased when N is increased. This phenomenon is observed in the simulations of Section IV as well. On the other hand, for the all-to-all network, the error bound is $\mathcal{O}(\sqrt{1/k})$ since $\lambda_2 = N$.

Remark 4: Theorem 1 also asserts that for any given $\epsilon > 0$, there is a sufficiently large k such that $\limsup_{t\to\infty} |x_i(t) - x_j(t)| \leq \epsilon$ for $i, j \in \mathcal{N}$. This property may be termed as 'practical (or, approximate) consensus' in the perspective of consensus control in [6], [14]–[16].

Proof: By the coordinate transformation

$$\xi = \begin{bmatrix}\xi_1\\ \tilde{\xi}\end{bmatrix} = Wx = \begin{bmatrix}\frac{1}{N}1_N^T\\ R^T\end{bmatrix}x$$

where $\tilde{\xi} = [\xi_2, \ldots, \xi_N]^T$, the overall system (7) is transformed into

$$\dot{\xi}_1 = \frac{1}{N}1_N^T f(t, 1_N\xi_1 + Q\tilde{\xi}) \tag{15a}$$

$$\dot{\tilde{\xi}} = -k\Lambda\tilde{\xi} + R^T f(t, 1_N\xi_1 + Q\tilde{\xi}) \tag{15b}$$

because $W^{-1}\xi = 1_N\xi_1 + Q\tilde{\xi}$. With $e := \xi_1 - s$, (15) can be rewritten as

$$\dot{e} = \frac{1}{N}1_N^T f(t, 1_N e + 1_N s + Q\tilde{\xi}) - \frac{1}{N}1_N^T f(t, 1_N s) \tag{16a}$$

$$\dot{\tilde{\xi}} = -k\Lambda\tilde{\xi} + R^T f(t, 1_N e + 1_N s + Q\tilde{\xi}). \tag{16b}$$

Let $V(e, \tilde{\xi}) := (1/2)e^2 + (1/2)|\tilde{\xi}|^2$. Then, the time derivative of V along (16) becomes

$$\begin{aligned}\dot{V} = \frac{e}{N}\Big[&1_N^T f(t, 1_N e + 1_N s + Q\tilde{\xi}) - 1_N^T f(t, 1_N e + 1_N s)\\ &+1_N^T f(t, 1_N e + 1_N s) - 1_N^T f(t, 1_N s)\Big]\\ &-k\tilde{\xi}^T\Lambda\tilde{\xi} + \tilde{\xi}^T R^T f(t, 1_N s)\\ &+\tilde{\xi}^T R^T\Big[f(t, 1_N e + 1_N s + Q\tilde{\xi}) - f(t, 1_N s)\Big]\end{aligned}$$

which leads, by the mean-value theorem, to

$$\begin{aligned}\dot{V} = \frac{e}{N}\left.\frac{\partial 1_N^T f}{\partial x}(t,x)\right|_z \cdot Q\tilde{\xi} + \frac{e}{N}\left.\frac{\partial 1_N^T f}{\partial e}(t, 1_N e)\right|_q \cdot e - k\tilde{\xi}^T\Lambda\tilde{\xi}\\ + \tilde{\xi}^T R^T\left.\frac{\partial f}{\partial x}(t,x)\right|_w \cdot (1_N e + Q\tilde{\xi}) + \tilde{\xi}^T R^T f(t, 1_N s)\end{aligned}$$

with some $z \in \mathbb{R}^N$, $w \in \mathbb{R}^N$, and $q \in \mathbb{R}$. Since

$$\tilde{\xi}^T R^T\left.\frac{\partial f}{\partial x}(t,x)\right|_w = \tilde{\xi}^T R^T \operatorname{diag}\left(\frac{\partial f_1}{\partial x_1}(t, w_1), \ldots, \frac{\partial f_N}{\partial x_N}(t, w_N)\right)$$

it is seen by (6) that

$$\left|\tilde{\xi}^T R^T\left.\frac{\partial f}{\partial x}(t,x)\right|_w\right| \leq L\|R\||\tilde{\xi}|, \quad \forall\, t \geq 0$$

and similarly that

$$\left|\left.\frac{\partial 1_N^T f}{\partial x}(t,x)\right|_z\right| \leq L\sqrt{N}, \quad \forall\, t \geq 0.$$

Therefore, using (9) and the fact that $\|Q\| = \sqrt{N}$ and $\|R\| = 1/\sqrt{N}$

$$\begin{aligned}\dot{V} \leq &-p|e|^2 + \frac{L\|Q\|}{\sqrt{N}}|e||\tilde{\xi}| - k\lambda_2|\tilde{\xi}|^2\\ &+ L\|R\||\tilde{\xi}|\left(\sqrt{N}|e| + \|Q\||\tilde{\xi}|\right)\\ &+ \left|R^T f(t, 1_N s)\right||\tilde{\xi}|\\ \leq &-p|e|^2 + 2L|e||\tilde{\xi}| - (k\lambda_2 - L)|\tilde{\xi}|^2 + |R^T f(t, 1_N s)||\tilde{\xi}|.\end{aligned}$$

Authorized licensed use limited to: Gwangju Institute of Science and Technology. Downloaded on November 10,2026 at 18:09:19 UTC from IEEE Xplore. Restrictions apply.

Lemma 2 is employed with $a := -L$, $\kappa := k\lambda_2 - L$, and $\theta(t) := |R^T f(t, 1_N s(t))|$, and we obtain that

$$\dot{V} \le -2cV \quad \text{if } 2V = e^2 + |\tilde{\xi}|^2 > |R^T f(t, 1_N s)|^2 r\left(\frac{1}{\kappa}\right) \tag{17}$$

where $c > 0$ and the function r is given by (14) (which can be seen by following the proof of Lemma 2 in the Appendix). Inequality (17) implies that

$$\limsup_{t\to\infty} 2V(t) \le \limsup_{t\to\infty} |R^T f\,(t, 1_N s(t))\,|^2 r\left(\frac{1}{\kappa}\right).$$

By (6) and Lemma 1, we have that

$$\limsup_{t\to\infty} \left|R^T f\,(t, 1_N s(t))\right| \le \|R\|\sqrt{N} M\left(\limsup_{t\to\infty} |s(t)|\right)$$
$$\le M\left(\frac{\limsup_{t\to\infty} |\bar{f}(t,0)|}{p}\right) \le M\left(\frac{M(0)}{p}\right).$$

Finally, since $x - 1_N s = W^{-1}\xi - 1_N s = 1_N\xi_1 - 1_N s + Q\tilde{\xi} = [1_N, Q][e, \tilde{\xi}]^T$, and the vector norm of the i-th row of $[1_N, Q]$ is $\sqrt{N}$ by the construction of W, we have $|x_i - s| \le \sqrt{N}\sqrt{|e|^2 + |\tilde{\xi}|^2} = \sqrt{N}\sqrt{2V}$. Therefore, for any $i \in \mathcal{N}$

$$\limsup_{t\to\infty} |x_i(t) - s(t)| \le M\left(\frac{M(0)}{p}\right)\sqrt{N}\sqrt{r\left(\frac{1}{\kappa}\right)}$$

if $\kappa = k\lambda_2 - L > 3L^2/p$. This completes the proof with (12) and (13). □

The proof of Theorem 1 enlightens the following. The quantity $|R^T f(t, 1_N s)|$ has the meaning of 'measure of heterogeneity' in the sense that, if all agents are identical; $f_i(t, s) = f_0(t, s)$ for all s and $i \in \mathcal{N}$, then $R^T f(t, 1_N s) = R^T 1_N f_0(t, s) = 0$. More specifically, if we denote the first column of R^T by r_1 so that $R^T = [r_1, \tilde{R}]$ with a matrix $\tilde{R}$, then it follows from $R^T 1_N = 0$ that $r_1 = -\tilde{R}1_{N-1}$. Hence

$$R^T f(t, 1_N s) = [r_1, \tilde{R}]\begin{bmatrix} f_1(t,s) \\ \vdots \\ f_N(t,s) \end{bmatrix} = \tilde{R}\begin{bmatrix} f_2(t,s) - f_1(t,s) \\ \vdots \\ f_N(t,s) - f_1(t,s) \end{bmatrix}.$$

In particular, if $R^T f(t, 1_N s) = 0$, then it can be seen that $\lim_{t\to\infty} |x_i(t) - x_j(t)| = 0$, $\forall\, i, j \in \mathcal{N}$, with some $k > 0$. While this can be seen directly from (17), it also follows from (15b) by appending $R^T f(t, 1_N\xi_1)$ (which is 0) and utilizing the Lipschitzness of f (Assumption 1). (Indeed, if $\tilde{\xi}(t) \to 0$ then $x_i(t) \to x_1(t)$, $\forall\, i$, since $\tilde{\xi} = R^T x = \tilde{R}[x_2 - x_1, \ldots, x_N - x_1]^T$.) The latter approach reveals that convergence of $\tilde{\xi}(t)$ to zero is in fact regardless of the behavior of $\xi_1(t)$. Under the contraction property [5] of f_0 (Assumption 3), the state $\xi_1(t)$ reaches a certain steady state forgetting the effect of the initial condition of $\xi_1(0)$ and $\tilde{\xi}(0)$, and this is in contrast to the 'average consensus' of [11], [14], [15], in which the agreed trajectory reflects the initial conditions of the networked agents.

III. Robustness by a Large Number of Agents

In this section, we are more specific on the system diversity and derive quantitative robustness measure under a large number of agents. For this, we introduce random variables to represent the heterogeneous group of N dynamic agents under consideration as

$$\dot{x}_i = f_i(t, x_i) + u_i = g_0(t, x_i) + \sum_{j=1}^{d} \Delta_{ij} g_j(t, x_i) + u_i \tag{18}$$

for $i \in \mathcal{N}$, where g_j's are C^1 "generating functions" for the individual agent function f_i, and Δ_{ij}'s are random variables having the expectation $E\{\Delta_{ij}\} = 0$ (without loss of generality) and the variance $V\{\Delta_{ij}\} = \sigma_j^2$ $(\sigma_j \ge 0)$ for all $i \in \mathcal{N}$ and $j \in \mathcal{D} := \{1, \ldots, d\}$.[3] It is assumed that Δ_{ij}'s are mutually independent for all i, but possibly dependent for j. The averaged dynamics of (18) is then given by

$$\dot{s} = \bar{f}(t, s) = \frac{1}{N}\sum_{i=1}^{N} f_i(t, s) = g_0(t, s) + \sum_{j=1}^{d} \bar{\Delta}_j g_j(t, s) \tag{19}$$

where $\bar{\Delta}_j := \sum_{i=1}^{N} \Delta_{ij}/N$. It should be noted that $E\{\bar{\Delta}_j\} = 0$ and $V\{\bar{\Delta}_j\} = \sigma_j^2/N$ [17]. Now let us consider the *expected averaged dynamics* which will be used as a reference for comparison

$$\dot{s}_E = E\left\{\bar{f}(t, s_E)\right\} = g_0(t, s_E), \quad s_E(0) = s(0). \tag{20}$$

Since we do not impose any particular conditions (such as probability distribution) on the random variables Δ_{ij}, they are unbounded, and thus, Assumptions 1 and 3 cannot hold for all cases. Instead, we impose assumptions on the generating functions.

Assumption 1:* There exist non-decreasing continuous functions $M_j : \mathbb{R}_{\ge 0} \to \mathbb{R}_{\ge 0}$ and constants $L_j > 0$ such that $|g_j(t, x_i)| \le M_j(|x_i|)$ and $|(\partial g_j/\partial x_i)(t, x_i)| \le L_j$ for all $x_i \in \mathbb{R}$, $t \ge 0$, $0 \le j \le d$.

Assumption 3:* There exists $p > 0$ such that $(\partial g_0/\partial s)(t, s) \le -p$ for all $s \in \mathbb{R}$ and $t \ge 0$.

To develop further, we recall a well-known fact for probability.

Lemma 3—(Chebyshev's Inequality): Let X be a random variable with $E\{X\} = \mu$ and $V\{X\} = \sigma^2$ $(\sigma \ge 0)$. Then, for any $h > 0$, it holds that

$$P\left(|X - \mu| \ge h\right) \le \frac{\sigma^2}{h^2}$$

and therefore, $1 - \sigma^2/h^2 \le P(|X - \mu| < h) \le P(|X - \mu| \le h)$.

Lemma 4: Let $A_1, \ldots, A_d$ be d arbitrary events. Then

$$\sum_{i=1}^{d} P(A_i) - d + 1 \le P\left(\bigcap_{i=1}^{d} A_i\right).$$

Proof: It is clearly true for $d = 1$. If it holds for $d = k - 1$, then

$$\sum_{i=1}^{k} P(A_i) - k + 1$$
$$= \sum_{i=1}^{k-1} P(A_i) - (k-1) + 1 + P(A_k) - 1$$
$$\le P\left(\cap_{i=1}^{k-1} A_i\right) + P(A_k) - 1 \le P\left(\left(\cap_{i=1}^{k-1} A_i\right) \cap A_k\right)$$
$$= P\left(\cap_{i=1}^{k} A_i\right)$$

in which, the second inequality follows from the fact that $P(A) + P(B) - P(A \cap B) \le 1$ for any two events A and B. □

Now we compare two trajectories $s(t)$ and $s_E(t)$ and claim that they tend to be arbitrarily close with high probability when N is sufficiently large. For this, note first that $\limsup_{t\to\infty} |s_E(t)| \le \limsup_{t\to\infty} |g_0(t, 0)|/p \le M_0(0)/p$, which follows from Lemma 1 for (20) under Assumptions 1* and 3*. Now let $\tilde{s} := s - s_E$. Then

$$\dot{\tilde{s}} = g_0\left(t, \tilde{s} + s_E(t)\right) - g_0\left(t, s_E(t)\right)$$
$$+ \sum_{j=1}^{d} \bar{\Delta}_j g_j\left(t, \tilde{s} + s_E(t)\right) =: \tilde{f}(t, \tilde{s})$$

[3] It is noted that (18) is not a stochastic differential equation. Once those random variables are drawn (or, realized) when the system is created, it remains deterministic. In this sense, we share the philosophy of "random differential equation" in [18].

Authorized licensed use limited to: Gwangju Institute of Science and Technology. Downloaded on November 10,2026 at 18:09:19 UTC from IEEE Xplore. Restrictions apply.

with $\tilde{s}(0)=0$. With this, suppose that

$$|\bar{\Delta}_j| \le \frac{p}{2dL_j} =: \phi_j, \quad \forall\, j \in \mathcal{D}. \tag{21}$$

Then, $(\partial\tilde{f}/\partial\tilde{s}) = (\partial\bar{f}/\partial s) = (\partial g_0/\partial s) + \sum_{j=1}^d \bar{\Delta}_j(\partial g_j/\partial s) \le -p + \sum_{j=1}^d |\bar{\Delta}_j| L_j \le -p/2$. Lemma 1 now yields that $\limsup_{t\to\infty}|\tilde{s}(t)| \le \limsup_{t\to\infty}|\tilde{f}(t,0)|/(p/2) = \limsup_{t\to\infty}|\sum_{j=1}^d \bar{\Delta}_j g_j(t, s_E(t))|/(p/2)$. Therefore, if, in addition to (21)

$$|\bar{\Delta}_j| \le \frac{\epsilon p}{2dM_j\left(\frac{M_0(0)}{p}\right)} =: \varphi_j, \quad \forall\, j \in \mathcal{D} \tag{22}$$

with any $\epsilon > 0$, then

$$\begin{aligned}\limsup_{t\to\infty}|\tilde{s}(t)| &\le \frac{\limsup_{t\to\infty}\sum_{j=1}^d |\bar{\Delta}_j|\,|g_j(t, s_E(t))|}{\frac{p}{2}} \\ &\le \frac{\limsup_{t\to\infty}\sum_{j=1}^d |\bar{\Delta}_j| M_j\left(\frac{M_0(0)}{p}\right)}{\frac{p}{2}} \le \epsilon.\end{aligned} \tag{23}$$

Finally

$$\begin{aligned}&P\left(|\bar{\Delta}_j| \le \phi_j \text{ and } |\bar{\Delta}_j| \le \varphi_j,\ \forall\, j \in \mathcal{D}\right) \\ &= P\left(\cap_{j=1}^d \left\{\left(|\bar{\Delta}_j| \le \phi_j\right) \cap \left(|\bar{\Delta}_j| \le \varphi_j\right)\right\}\right) \\ &\ge \sum_{j=1}^d P\left(\left(|\bar{\Delta}_j| \le \phi_j\right) \cap \left(|\bar{\Delta}_j| \le \varphi_j\right)\right) - d + 1 \text{ by Lemma 4} \\ &\ge \sum_{j=1}^d P\left(|\bar{\Delta}_j| \le \phi_j\right) P\left(|\bar{\Delta}_j| \le \varphi_j\right) - d + 1\end{aligned}$$

since one event implies the other and probability is not larger than 1, and Lemma 3 leads to

$$\ge \sum_{j=1}^d \left(1 - \frac{1}{\phi_j^2}\frac{\sigma_j^2}{N}\right)\left(1 - \frac{1}{\varphi_j^2}\frac{\sigma_j^2}{N}\right) - d + 1. \tag{24}$$

In summary, we conclude that

$$\begin{aligned}\sum_{j=1}^d \left(1 - \frac{4d^2\sigma_j^2 L_j^2}{p^2 N}\right)\left(1 - \frac{4d^2\sigma_j^2 M_j\left(\frac{M_0(0)}{p}\right)^2}{\epsilon^2 p^2 N}\right) - d + 1 \\ \le P\left(\limsup_{t\to\infty}|s(t) - s_E(t)| \le \epsilon\right).\end{aligned} \tag{25}$$

Note that this also implies (5).

The following theorem characterizes the effect of a large number of agents and is the second main result of the technical note.

Theorem 2: Under Assumptions 1*, 2, and 3*, there exists a class-$\mathcal{K}$ function γ_N^* such that the solutions of the overall system, composed of (18) and (2), satisfy that

$$\begin{aligned}P\left(\limsup_{t\to\infty}|x_i(t) - s_E(t)| \le \gamma_N^*\left(\frac{1}{k\lambda_2 - \mathcal{L}_N}\right)\right. \\ \left. + \epsilon, \forall\, k > \bar{\mathcal{K}}_N,\ \forall\, i \in \mathcal{N}\right) \ge \left(1 - \frac{d}{N^2}\right)^N \\ + \sum_{j=1}^d \left(1 - \frac{4d^2\sigma_j^2 L_j^2}{p^2 N}\right)\left(1 - \frac{4d^2\sigma_j^2 M_j\left(\frac{M_0(0)}{p}\right)^2}{\epsilon^2 p^2 N}\right) - d\end{aligned} \tag{26}$$

where $s_E(t)$ is the solution of (20) with $s_E(0) = \sum_{i=1}^N x_i(0)/N$, ϵ is any positive number, and

$$\mathcal{L}_N := L_0 + N\sum_{j=1}^d \sigma_j L_j \quad \text{and} \quad \bar{\mathcal{K}}_N := \frac{6\mathcal{L}_N^2}{p\lambda_2} + \frac{\mathcal{L}_N}{\lambda_2}. \tag{27}$$

The function γ_N^* is defined on $[0, p/6\mathcal{L}_N^2)$ and given by

$$\gamma_N^*(\chi) = \mathcal{M}_N\left(\frac{2\mathcal{M}_N(0)}{p}\right)\sqrt{N}\sqrt{r_N(\chi)} \tag{28}$$

in which

$$\mathcal{M}_N(a) := M_0(a) + N\sum_{j=1}^d \sigma_j M_j(a), \quad \text{and}$$

$$r_N(\chi) := \begin{cases} 0, & \chi = 0 \\ \frac{8\chi}{p - 6\mathcal{L}_N^2\chi}, & 0 < \chi \le \frac{8p}{p^2 + 80\mathcal{L}_N^2} \\ \frac{\left(p^2 + 32\mathcal{L}_N^2\right)\chi^2}{\left(p - 6\mathcal{L}_N^2\chi\right)^2}, & \frac{8p}{p^2+80\mathcal{L}_N^2} < \chi < \frac{p}{6\mathcal{L}_N^2}. \end{cases} \tag{29}$$

A few factors are related to the upper bound of $\limsup_{t\to\infty}|x_i(t) - s_E(t)|$ and the probability estimation in (26). The upper bound tends to increase as N increases since $\mathcal{L}_N$ and $\mathcal{M}_N$ increase. (Also, λ_2 is affected.) The lower bound of the probability [the right-hand side of (26)] tends to decrease as the increment of d, σ_j, L_j, M_j, and the decrement of p, which intuitively makes sense. It is also true that, with ϵ getting larger, the lower bound of the probability increases. In any case, as $k \to \infty$, the upper error bound in (26) tends to ϵ, and the probability of achieving this error bound tends to 1 as $N \to \infty$. The latter can be seen from the right-hand side of (26), with

$$\lim_{N\to\infty}\left(1 - \frac{d}{N^2}\right)^N = \lim_{N\to\infty}\left\{\left(1 - \frac{d}{N^2}\right)^{-\frac{N^2}{d}}\right\}^{-\frac{d}{N}} = 1.$$

Therefore, it is asserted that, with a large number of agents, it is more likely to have $\limsup_{t\to\infty}|x_i(t) - s_E(t)|$ less than or equal to the error bound for all $i \in \mathcal{N}$, and the error bound can be made small with strong coupling k.

Remark 5: As in Remark 3, let us look into the ultimate error bound γ_N^* in more detail. For this, suppose that, without loss of generality, the function M_j in Assumption 1* is taken as an affine function of $|x_i|$ (for example, $M_j(|x_i|) = \sup_{t\ge 0}|g_j(t,0)| + L_j|x_i|$ does the job). Then, with sufficiently large k, one can show that $\gamma_N^*(1/(k\lambda_2 - \mathcal{L}_N)) = \mathcal{O}(\sqrt{N^5/k\lambda_2})$.

Proof: In this proof, we combine the analysis performed for Theorem 1 which discusses the closeness between $x_i(t)$ and $s(t)$, and the analysis about the closeness between $s(t)$ and $s_E(t)$. For this, let us first suppose that

$$|\Delta_{ij}| \le N\sigma_j, \quad \forall\, i \in \mathcal{N},\ \forall\, j \in \mathcal{D}. \tag{30}$$

Then, it follows from Assumption 1* that $|f_i(t,x_i)| = |g_0(t,x_i) + \sum_{j=1}^d \Delta_{ij} g_j(t,x_i)| \le M_0(|x_i|) + N\sum_{j=1}^d \sigma_j M_j(|x_i|) \le \mathcal{M}_N(|x_i|)$, and that $|(\partial f_i/\partial x_i)(t,x_i)| = |(\partial g_0/\partial x_i) + \sum_{j=1}^d \Delta_{ij}(\partial g_j/\partial x_i)| \le L_0 + N\sum_{j=1}^d \sigma_j L_j = \mathcal{L}_N$, so that Assumption 1 holds with M and L replaced with $\mathcal{M}_N$ and $\mathcal{L}_N$, respectively. Moreover, if (21) holds as well, then $|(\partial\bar{f}/\partial s)| \le -p/2$ (not $-p$!) as seen before. In this case, Theorem 1 guarantees that

$$\limsup_{t\to\infty}|x_i(t) - s(t)| \le \gamma_N^*\left(\frac{1}{k\lambda_2 - \mathcal{L}_N}\right),\ \forall\, k > \bar{\mathcal{K}}_N,\ \forall\, i \in \mathcal{N}$$

in which, γ_N^*, $\bar{\mathcal{K}}_N$, and r_N are given by (28), (27), and (29), respectively (which are obtained simply by replacing p in Theorem 1 with $p/2$). Now, if (22) holds additionally with some $\epsilon > 0$, then we have seen that $\limsup_{t\to\infty}|s(t) - s_E(t)| \le \epsilon$ in (23). Thus, combining

Authorized licensed use limited to: Gwangju Institute of Science and Technology. Downloaded on November 10,2026 at 18:09:19 UTC from IEEE Xplore. Restrictions apply.

these two by triangular inequality, we obtain that $\limsup_{t\to\infty}|x_i(t)-s_E(t)|\le\gamma_N^*(1/(k\lambda_2-\mathcal{L}_N))+\epsilon$ for all $k>\bar{\mathcal{K}}_N$ and $i\in\mathcal{N}$.

To see (26), it is left to compute the probability that all (21), (22), and (30) hold. For (30), it is seen that

$$\begin{aligned}
&P\left(|\Delta_{ij}|\le N\sigma_j,\ \forall\, i\in\mathcal{N},\ \forall\, j\in\mathcal{D}\right)\\
&=\prod_{i=1}^{N}P(|\Delta_{ij}|\le N\sigma_j,\forall\, j\in\mathcal{D}) \text{ by independency of } \Delta_{ij} \text{ for } i\\
&\ge\prod_{i=1}^{N}\left(\sum_{j=1}^{d}P\left(|\Delta_{ij}|\le N\sigma_j\right)-d+1\right) \text{ by Lemma 4}\\
&\ge\prod_{i=1}^{N}\left(\sum_{j=1}^{d}\left(1-\frac{1}{N^2}\right)-d+1\right) \quad \text{by Lemma 3}\\
&=\left(1-\frac{d}{N^2}\right)^N.
\end{aligned}$$

Finally, we have that

$$\begin{aligned}
&P(((30)\text{ holds})\cap((21)\text{ and }(22)\text{ hold}))\\
&\quad\ge P\left((30)\text{ holds}\right)+P\left((21)\text{ and }(22)\text{ hold}\right)-1\\
&\quad\ge\left(1-\frac{d}{N^2}\right)^N+\sum_{j=1}^{d}\left(1-\frac{1}{\phi_j^2}\frac{\sigma_j^2}{N}\right)\left(1-\frac{1}{\varphi_j^2}\frac{\sigma_j^2}{N}\right)-d \text{ from } (24)
\end{aligned}$$

which concludes (26). □

IV. Simulations

As an example, consider a group of N agents with

$$\begin{aligned}
f_i(t,x_i)=(-1+\delta_i)x_i+10\sin t\\
+10m_i^1\sin\left(0.1t+\theta_i^1\right)+10m_i^2\sin\left(10t+\theta_i^2\right)
\end{aligned}$$

for $i=1,2,\ldots,N$, where δ_i, m_i^1, and m_i^2 are realizations of independent random variables of standard normal distribution $N(0,1)$, and θ_i^1 and θ_i^2 are realizations of independent random variables of uniform distribution on $[0,2\pi]$. Since, for example

$$\begin{aligned}
10m_i^1\sin\left(0.1t+\theta_i^1\right)=10m_i^1\cos\theta_i^1\sin 0.1t\\
+10m_i^1\sin\theta_i^1\cos 0.1t
\end{aligned}$$

in order to represent the agent in the form of (18), we take $\Delta_{i1}=\delta_i$, $\Delta_{i2}=10m_i^1\cos\theta_i^1$, $\Delta_{i3}=10m_i^1\sin\theta_i^1$, $\Delta_{i4}=10m_i^2\cos\theta_i^2$, and $\Delta_{i5}=10m_i^2\sin\theta_i^2$ with $g_0(t,x_i)=-x_i+10\sin t$, $g_1(t,x_i)=x_i$, $g_2=\sin 0.1t$, $g_3=\cos 0.1t$, $g_4=\sin 10t$, and $g_5=\cos 10t$. Indeed, $E\{\Delta_{ij}\}=0$ for all i and j since m_i^j and θ_i^j are independent and $E\{m_i^j\}=0$. Also, Δ_{ij} are independent for i while it is not for j. Assuming that these agents are interconnected by the ring network, the simulation results of two sample runs with $N=5$ and $N=100$ are depicted in Fig. 1. In each case, the coupling gains of $k=5$ and $k=500$ are used, respectively.

The effect of strong coupling k is seen rather clearly by comparing Fig. 1(a) with (b), and (c) with (d), respectively. On the other hand, by comparing (a) with (c), and (b) with (d), it is seen that, when N is larger, the solutions of each agents (solid blue) tend closer to the dashed black curve, which is the solution of the expected averaged dynamics $\dot{s}_E=-s_E+10\sin t$, $s_E(0)=(1/N)\sum_{i=1}^N x_i(0)$. This is because the averaged dynamics, given by $\dot{s}=(-1+(1/N)\sum_{i=1}^N\delta_i)s+10\sin t+(1/N)\sum_{i=1}^N 10m_i^1\sin(0.1t+\theta_i^1)+(1/N)\sum_{i=1}^N 10m_i^2\sin(10t+\theta_i^2)$, gets close to the expected averaged dynamics with large N. In the sample run for $N=5$, it was $\dot{s}=-0.437s+10\sin t+0.445\sin 0.1t+6.005\cos 0.1t+6.588\sin 10t-5.523\cos 10t$, and $\dot{s}=-1.034s+10\sin t+0.608\sin 0.1t-0.833\cos 0.1t-0.671\sin 10t+0.08\cos 10t$ for $N=100$.

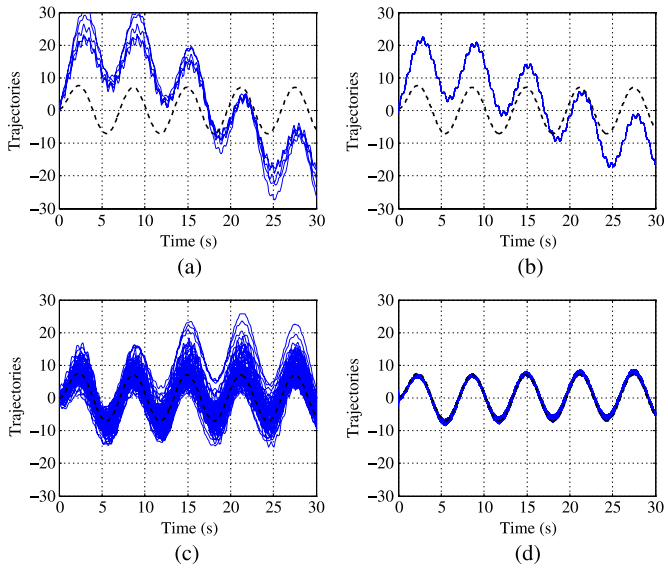


Fig. 1. Trajectories of N-agent systems with coupling strength k are depicted as blue solid curves, and the trajectory $s_E(t)$ of the expected averaged system is given as the black dashed curve. (a) $N=5$, $k=5$. (b) $N=5$, $k=500$. (c) $N=100$, $k=5$. (d) $N=100$, $k=500$.

It is also observed that increasing N (under the same k) results in more deviation by comparing (a) and (c); (b) and (d), as discussed in Remark 3. Finally, it is stressed that there were 2 unstable agents out of 5 when $N=5$, and 12 out of 100 when $N=100$, in the sample run. By repeating the sample runs, we observe that the error $|x_i(t)-x_j(t)|$ is small for all $i,j\in\mathcal{N}$ when k is large, and the deviation of $x_i(t)$ from $s_E(t)$ is small *with high probability* when both N and k are large. Fig. 1 is a typical one of them.

V. Further Discussion

We have claimed that strong coupling and a large number of agents both enhance robustness of the networked group behavior. While strong coupling enforces the behavior of each agent tend to that of averaged dynamics, a large number of agents attenuate the effects of stochastic uncertainty in the averaged dynamics. Mathematical analysis of the robustness by strong coupling and a large number of agents is provided and this phenomenon is illustrated by simulation. The role of the averaged dynamics of heterogeneous (possibly uncertain) agents has been emphasized, with which trading stability among the agents by interconnection is seen rather clearly. As a result, even unstable agents are allowed in the network as long as their instability is compensated by other's stability through the network.

While only the scalar systems are dealt with in this technical note, extensions to higher order systems seem necessary for further study.

Appendix

Lemma 2 is proved as follows. With $\kappa>3a^2/p$, note that

$$\begin{aligned}
&\rho_\kappa(x,y)+\frac{p}{2}|x|^2+\frac{a^2}{p}|y|^2\\
&=-p|x|^2-2a|x||y|-\kappa|y|^2+\theta(t)|y|+\frac{p}{2}|x|^2+\frac{a^2}{p}|y|^2\\
&=-\frac{p}{2}\left(|x|+\frac{2a}{p}|y|\right)^2-\left(\kappa-\frac{3a^2}{p}\right)|y|^2+\theta(t)|y|\\
&=-\frac{p}{2}X^2-\delta(\kappa)|y|^2+\theta(t)|y|\\
&=-\delta(\kappa)\left(\frac{p}{2\delta(\kappa)}X^2+|y|^2\right)+\theta(t)|y|\\
&\le-\delta(\kappa)|Y_\kappa|^2+\theta(t)|Y_\kappa|
\end{aligned}$$

Authorized licensed use limited to: Gwangju Institute of Science and Technology. Downloaded on November 10,2026 at 18:09:19 UTC from IEEE Xplore. Restrictions apply.

where $X := |x| + (2a/p)|y|$, $\delta(\kappa) := \kappa - 3a^2/p$, and $Y_\kappa := [\sqrt{p/2\delta(\kappa)}X, |y|]^T$. The last inequality holds because $|Y_\kappa| \geq |y|$. Therefore, it follows that if $|Y_\kappa| \geq \theta(t)/\delta(\kappa)$, then $\rho_k(x,y) \leq -c(|x|^2 + |y|^2)$ with $c := \min\{p/2, a^2/p\}$. Now, it is seen that

$$\begin{aligned}|x|^2 + |y|^2 &= \left(X - \frac{2a}{p}|y|\right)^2 + |y|^2 \leq 2X^2 + \frac{8a^2}{p^2}|y|^2 + |y|^2 \\ &\leq \max\left\{\frac{4\delta(\kappa)}{p}, \frac{8a^2}{p^2} + 1\right\}\left(\frac{p}{2\delta(\kappa)}X^2 + |y|^2\right) \\ &=: \eta(\kappa)|Y_\kappa|^2.\end{aligned}$$

With $\eta(\kappa)/\delta^2(\kappa)$ being monotonically decreasing to zero as $\kappa \to \infty$ for $\kappa > 3a^2/p$, define a class-$\mathcal{K}$ function $r : [0, p/3a^2) \to [0, \infty)$ as follows:

$$\begin{aligned}r(\chi) &= \begin{cases} 0, & \chi = 0 \\ \frac{\eta\left(\frac{1}{\chi}\right)}{\delta^2\left(\frac{1}{\chi}\right)}, & 0 < \chi < \frac{p}{3a^2} \end{cases} \\ &= \begin{cases} 0 & \chi = 0 \\ \frac{4\chi}{p - 3a^2\chi}, & 0 < \chi \leq \frac{4p}{p^2 + 20a^2} \\ \frac{(p^2 + 8a^2)\chi^2}{(p - 3a^2\chi)^2}, & \frac{4p}{p^2 + 20a^2} < \chi < \frac{p}{3a^2} \end{cases}\end{aligned}$$

in which, if $a = 0$, the number $p/3a^2$ is considered as ∞.

Therefore, if $|x|^2 + |y|^2 > \theta^2(t) r(1/\kappa)$, then

$$\begin{aligned}|Y_\kappa|^2 &\geq \frac{1}{\eta(\kappa)}\left(|x|^2 + |y|^2\right) > \frac{\theta^2(t)}{\eta(\kappa)} r\left(\frac{1}{\kappa}\right) \geq \frac{\theta^2(t)}{\eta(\kappa)} \frac{\eta(\kappa)}{\delta^2(\kappa)} \\ &= \left(\frac{\theta(t)}{\delta(\kappa)}\right)^2\end{aligned}$$

which completes the proof.

REFERENCES

[1] J. Garcia-Ojalvo, M. B. Elowitz, and S. H. Strogatz, "Modeling a synthetic multicellular clock: Repressilators coupled by quorum sensing," *Proc. National Academy of Sciences*, vol. 101, no. 30, pp. 10955– 10960, Jul. 2004.

[2] J. Kim, P. Heslop-Harrison, I. Postlethwaite, and D. Bates, "Stochastic noise and synchronisation during dictyostelium aggregation make camp oscillations robust," *PLoS Comput. Biol.*, vol. 3, no. 11, pp. 2190– 2198, 2007.

[3] N. Tabareau, J. Slotine, and Q. Pham, "How synchronization protects from noise," *PLoS Comput. Biol.*, vol. 6, no. 1, pp. 1– 9, 2010.

[4] J. Hale, "Diffusive coupling, dissipation, and synchronization," *J. Dynam. Diff. Equat.*, vol. 9, no. 1, pp. 1– 52, 1997.

[5] W. Lohmiller and J. Slotine, "On contraction analysis for nonlinear systems," *Automatica*, vol. 34, no. 6, pp. 683– 696, 1998.

[6] J. Zhao, D. Hill, and T. Liu, "Synchronization of dynamical networks with nonidentical nodes: Criteria and control," *IEEE Trans. Circuits Syst.*, vol. 58, no. 3, pp. 584– 594, Mar. 2011.

[7] J. Zhao, D. Hill, and T. Liu, "Global bounded synchronization of general dynamical networks with nonidentical nodes," *IEEE Trans. Autom. Control*, vol. 57, no. 10, pp. 2656– 2662, Oct. 2012.

[8] D. A. Paley, N. E. Leonard, R. Sepulchre, D. Grünbaum, and J. K. Parrish, "Oscillator models and collective motion," *IEEE Control Syst. Mag.*, pp. 89– 105, Aug. 2007.

[9] J. Bouvrie and J. Slotine, "Synchronization and redundancy: Implications for robustness of neural learning and decision making," *Neural Comput.*, vol. 23, no. 11, pp. 2915– 2941, 2011.

[10] Q. Pham and J. Slotine, "Stable concurrent synchronization in dynamic system networks," *Neural Netw.*, vol. 20, pp. 62– 77, 2007.

[11] R. Olfati-Saber and J. Shamma, "Consensus filters for sensor networks and distributed sensor fusion," in *Proc. Joint 44th IEEE Conf. Decision Control European Control Conf.*, 2005, pp. 6698– 6703.

[12] H. Khalil, *Nonliner Systems*, 3rd ed. Upper Saddle River, NJ, USA: Prentice-Hall, 2002.

[13] M. Fiedler, "Algebraic connectivity of graphs," *Czech. Mathemat. J.*, vol. 23, no. 98, pp. 298– 305, 1973.

[14] R. Olfati-Saber, J. Fax, and R. Murray, "Consensus and cooperation in networked multi-agent systems," *Proc. IEEE*, vol. 95, no. 1, pp. 215– 233, Jan. 2007.

[15] J. H. Seo, H. Shim, and J. Back, "Consensus of high-order linear systems using dynamic output feedback compensator: Low gain approach," *Automatica*, vol. 45, no. 11, pp. 2659– 2664, 2009.

[16] H. Kim, H. Shim, and J. H. Seo, "Output consensus of heterogeneous uncertain linear multi-agent systems," *IEEE Trans. Autom. Control*, vol. 56, no. 1, pp. 200– 206, Jan. 2011.

[17] R. D. Yates and D. J. Goodman, *Probability and Stochastic Processes*. New York, NY, USA: Wiley, 1998.

[18] N. Tabareau and J. Slotine, "Contraction analysis of nonlinear random dynamical systems," arXiv preprint arXiv:1309.5317, 2013.

[19] J. Kim, J. Yang, H. Shim, and J.-S. Kim, "Robustness of synchronization in heterogeneous multi-agent systems," in *Proc. 12th European Control Conf.*, 2013, pp. 3821– 3826.

[20] P. DeLellis, M. Bernardo, and D. Liuzza, "Convergence and synchronization in heterogeneous networks of smooth and piecewise smooth systems," *Automatica*, vol. 56, no. 6, pp. 1– 11, 2015.

Authorized licensed use limited to: Gwangju Institute of Science and Technology. Downloaded on November 10,2026 at 18:09:19 UTC from IEEE Xplore. Restrictions apply.